

Checklist for Development and Report Writing

When you are doing the work, when you are writing and when you think you are finished with your work and the report, you should go through this checklist to see if everything is OK before you deliver. This checklist gives guidelines for the work and the delivered documentation/report. If any of the items in the checklist are not OK, make sure to improve before delivery. In addition to the checklist, an overall assessment will be carried out to see if you have done enough to pass the assignment.

#	Item	OK
1	Think about the Assignments as a small real-life industrial projects that you perform on behalf of a customer or client, and not a set of tasks, exercises or a "School Assignment".	
2	Try to see your work in a larger context than just an Assignment or a set of exercises. Try to see the big picture. The tasks within the assignment are just small building blocks that ends up with a fully working system.	
3	You have included a separate Title page with a Title and your Name. Typically, a Date is also appropriate to include.	
4	You have included a Table of Contents in the document.	
5	You have NOT included a Page number on the Title page. The Title page should be without page number.	
6	You have included Page numbers in the rest of the report (typically centered in the footer). If you have an Introduction chapter, that is typically page 1, while pages before the Introduction chapter typically use i, ii, ..etc.	
7	If you have included an Image on the title page (which is optional), the image should be relevant and in context of your work, not just a general image found on Internet or made by AI. If you use images that is not yours, you need to include a reference.	
8	Your Headings/Chapters are using numbering , e.g., "1 Heading1", "1.1 Heading2", etc. The Introduction chapter is always chapter 1 .	
9	You have included a System Sketch . The System Sketch is a figure that gives the overview of what you have done, i.e., the "Big Picture". Make System Sketch in PowerPoint: https://youtu.be/9mmBXFOjV3s	
10	You have started each Chapter and Subchapter with a short introduction text before you present any Figures, Tables, a list of bullet points, etc.	
11	You have NOT used any Figures, Tables or directly copied Equations from the Assignment. Make sure to make your own Figures, Sketches, Tables, etc. where you show how you understand it and presenting the work (not others).	
12	You have NOT used the words like "I", "My" or "We" in the text , meaning you have NOT written like this "In my application I have implemented a PID controller using LabVIEW...", but I have written something like this "In the application a PID controller has been implemented using LabVIEW...".	
13	Figure Captions : For each Figure you have added a Figure number and Figure Title below the Figure, e.g., "Figure 2-3: Overview of Control System". It also looks better when the Figure is centered.	

14	Referring to Figures in the Text: For each Figure you have referred to that Figure in the text, e.g., "In Figure 2-3 we see the control system developed in this project. The control system consists of a PID controller..."	
15	For each Table you have added a Table number and Table title ABOVE the Table, e.g., "Table 3-4: PID Parameters for selected Tuning methods"	
16	For each Table you have referred to that Table in the text, e.g., "In Table 3-4 we see the PID parameters for the different tuning methods used in this project, these tuning methods...."	
17	The Equations are centered and have an Equation number (Note! The chapter number should be included as part of the equation number) that is right centered, e.g., $y = ax + b \quad (2-1)$ Equations in Microsoft Word: https://youtu.be/b9f2bb2yn1Y	
18	For each Equation you have referred to that Equation in the text, e.g., "From eq. (2-1) we see the linear relationship between the input signal and the output signal".	
19	You have NOT copied any Equations from the Assignments and passed them in as a Figure my report. Make your own equations if needed.	
20	You have NOT used multiplication sign "*" in equations (e.g., $y = a * x + b$). You don't use that when typing equations with pen and paper, so you don't need to use it in a report either. See how it is done I different textbooks. This looks better: $y = ax + b$ (no "*" is used)	
21	In equations you typically use letters and not "words", e.g., K_p for pump gain, NOT k_{pump} or something. Then, in the text below the formula/equation you explain what K_p is with a proper unit.	
22	In formulas and equations try to use upper case for constants (e.g. G or F) and lower case for variables (e.g., x , y , u) (but no rule without exception).	
23	You have NOT used words/sentences like "I am a student...", "In this assignment we shall...", "In task 4 we are supposed to do..."	
24	You have included Units in all my plots/charts, both on the x-axis and on the y-axis, this yields for plots/charts created in LabVIEW, C# but also for plots/charts created in Excel, etc. Assume you build a bridge and do not care if you use 100 kg or 100 ton in your calculations?	
25	You have included Units in the GUI, e.g., for input fields for T_i or when presenting, e.g., a temperature value $T = 20^\circ\text{C}$ Assume you build a bridge and do not care if you use 100 kg or 100 ton in your calculations?	
26	You have included Units when presenting values and doing calculations inside the report/lab summary, e.g., $T_i = 20\text{s}$	
27	Number of decimals: You have NOT presented values from e.g., a temperature sensor with 4+ decimals in my GUI or inside the report since this makes no sense because a temperature sensor is not that accurate. You have checked the datasheet for the sensor you are using.	

28	The GUIs are well structured and intuitive, e.g., the “Stop/Exit” button is placed in the lower right corner, elements in the GUI are logical structured, etc. LabVIEW Guidelines: https://youtu.be/rpQUO5isCbE	
29	The GUI screenshots inside the report show some proper/realistic data. In that way it is much easier to understand/see the purpose with the GUI.	
30	The LabVIEW code flows from left to right, i.e., the LabVIEW code doesn’t look like “spaghetti”. LabVIEW Programming Guidelines: https://youtu.be/rpQUO5isCbE	
31	The LabVIEW State Machine is a good method for structuring the LabVIEW code. It works for small applications as well. LabVIEW State Machine video: https://youtu.be/AIDivLw0slo	
32	You have created and used SubVIs in the LabVIEW code, you have also created an “Icon” since it takes just a few seconds to make a simple icon just using words or letters like this:  and it makes it so much easier to maintain and understand the code.	
33	You are not using any “strong” colors except for e.g., alarm handling or other situations that require “strong” colors.	
34	Buttons: Since the button has a text inside saying “Stop”, you have hidden the button label “Stop Button” that is by default is placed above the button. 	
35	You have used proper names and labeling for files (NOT like “Test1.vi”, “Peters PID Controller.vi”), variables (NOT “Numeric Control” but e.g., “Temperature”), user interface objects (NOT “Waveform Chart” but e.g., “Temperature Chart”), etc.	
36	You have spelled LabVIEW correct, NOT “LABVIEW”, Lab View”, “Labview”, etc. This yields also for other nouns, product name, corporate names, etc.	
37	Discussions: The results of your work are Discussed , e.g., “The Skogestad tuning gives better control performance than the Ziegler-Nichols method when used in the simulator...” and/or something like this: “The results from the simulations given in Table 3-4 shows that the control system works fine when applying a step response. The performance is also good when applying noise to the signal”	
38	Make sure to see the “ Big picture ”, meaning you have not focused on unnecessary details or included very basic stuff, nor are you talking about “Task 1”, “Task 2”, etc.	
39	You have included a Conclusion since a (short) conclusion is always needed. The conclusion should make sense and provide useful information to the reader regarding the technical work that has been done. You have shortly and precisely summarized your results and drawn conclusions. You have NOT written how much you have learned, or saying things like “This lab assignment was fun”, “This will be useful when I get a job”, etc.	
40	References have been included if you use information from other sources than the assignment or information provided by the supervisor. In addition to the Reference list itself, you have inside the report where the source is	

	used referred to the reference using a number, e.g., [2], e.g., “From [2] we know that there is a linear relationship between the voltage and the temperature value in degrees Celsius.”	
41	You have read the entire report and have found no obvious mistakes, spelling mistakes , it has a clear red thread, a good structure, etc.	
42	Note! The text should primarily be reflective and not descriptive. The text should not merely describe what has been done, but also why and how and what the results are. The outcome needs to be put into relevant context.	
43	You have solved (or at least tried) the main parts/tasks of the assignment, and you have also addressed those in the report.	
44	You have also “ Added Value ” compared to the simplified examples given. This means you have done the “little extra” that makes your solution “stand out” and that it has a “personal” touch.	
45	You have a good Report structure . You should use the IMRaD structure. IMRaD is an acronym for Introduction – Methods – Results – and – Discussions . The IMRaD format is a way of structuring a scientific article, the chapters within may have different names.	
46	The complementary Quiz has been taken and the score was above the minimum level	
47	After you have generated the final PDF file, make sure to open it and read through the entire text and make sure that you no find obvious mistakes, spelling mistakes, etc. You have also checked that there are none “Reference not found...”, etc.	

Additional Resources:

Report:

Report Template and Guidelines:

<https://www.halvorsen.blog/documents/teaching/courses/industrialit/resources/Report%20Template%20and%20Guidlines.docx>

Write Technical Reports in Microsoft Word: https://youtu.be/ao_eDJOEUkA

Make System Sketch in PowerPoint: <https://youtu.be/9mmBXFOjV3s>

Figures and Equations in Microsoft Word and PowerPoint:

<https://youtu.be/b9f2bb2yn1Y>

Citation and Referencing with Microsoft Word: https://youtu.be/IgH7qmLa_L4

LabVIEW/GUI:

LabVIEW Fundamentals:

https://halvorsen.blog/documents/programming/labview/labview_fundamentals.php

LabVIEW Programming Guidelines: <https://youtu.be/rpQUO5isCbE>

LabVIEW State Machine: <https://youtu.be/-b9St8wNhpQ>